INCH-POUND

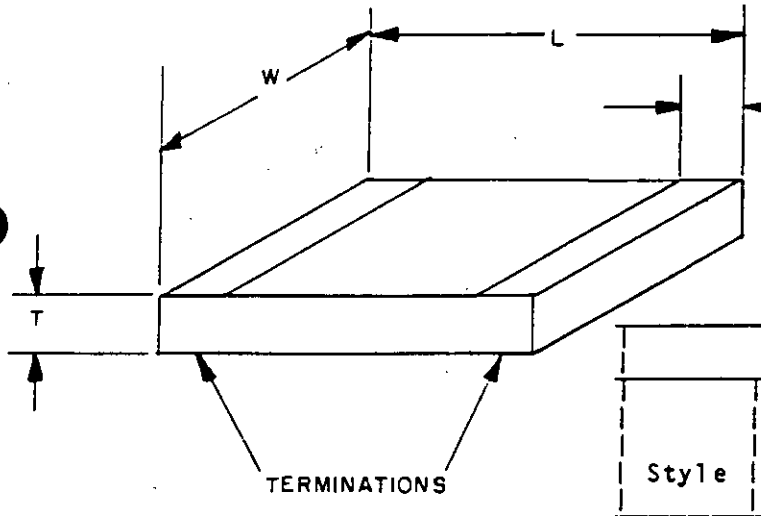
MIL-C-55681/1B
24 July 1990
SUPERSEDING
MIL-C-55681/1A
21 March 1972

MILITARY SPECIFICATION SHEET

CAPACITORS, CHIP, MULTIPLE LAYER, FIXED, UNENCAPSULATED, CERAMIC DIELECTRIC, ESTABLISHED RELIABILITY, STYLES CDR01, CDR02, CDR03, AND CDR04

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55681.



Inches	mm
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.030	0.76
.050	1.27
.055	1.40
.080	2.03
.125	3.18
.180	4.57

Dimensions				
			T	
Style	L ±.015	W ±.015	Min	Max
CDR01	.080	.050	.022	.055
CDR02	.180	.050	.022	.055
CDR03	.180	.080	.022	.080
CDR04	.180	.125	.022	.080

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
Ⓑ 3. For style CDR01, termination width shall be 0.010 inches, minimum; uncovered band width shall be 0.030 inches, minimum.
Ⓑ 4. Dimensions and tolerances are for bare chips. For solder coated terminations, for style CDR01 add .020 inch (0.51 mm) to the positive length tolerance and .015 inch (0.38 mm) to the positive width and thickness tolerance. For solder coated terminations for styles CDR02, CDR03, and CDR04, add .025 inch (0.64 mm) to the positive length tolerance and .015 inch (0.38 mm) to the positive width and thickness tolerance.

FIGURE 1. Styles CDR01, CDR02, CDR03, and CDR04 capacitors.

Ⓑ denotes changes

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Case type: Multi-layer, unencapsulated, monolithic.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55°C to +125°C.

Marking: In accordance with MIL-C-55681.

TABLE I. CDR01, CDR02, CDR03, and CDR04 capacitor characteristics.

Type designation 1/	Capacitance	Capacitance tolerance	Rated temperature and voltage temperature limits	Rated voltage
	pF			volts, dc
CDR01BP100B	10	J, K	BP	100
CDR01BP120BJ	12	J	BP	"
CDR01BP150B	15	J, K	BP	"
CDR01BP180BJ	18	J	BP	"
CDR01BP220B	22	J, K	BP	"
CDR01BP270BJ	27	J	BP	"
CDR01BP330B	33	J, K	BP	"
CDR01BP390BJ	39	J	BP	"
CDR01BP470B	47	J, K	BP	"
CDR01BP560BJ	56	J	BP	"
CDR01BP680B	68	J, K	BP	"
CDR01BP820BJ	82	J	BP	"
CDR01BP101B	100	J, K	BP	"
CDR01B-121B	120	J, K	BP, BX	"
CDR01B-151B	150	J, K	BP, BX	"
CDR01B-181B	180	J, K	BP, BX	"
CDR01BX221B	220	K, M	BX	"
CDR01BX271BK	270	K	BX	"
CDR01BX331B	330	K, M	BX	"
CDR01BX391BK	390	K	BX	"
CDR01BX471B	470	K, M	BX	"
CDR01BX561BK	560	K	BX	"
CDR01BX681B	680	K, M	BX	"
CDR01BX821BK	820	K	BX	"
CDR01BX102B	1,000	K, M	BX	"
CDR01BX122BK	1,200	K	BX	"
CDR01BX152B	1,500	K, M	BX	"
CDR01BX182BK	1,800	K	BX	"
CDR01BX222B	2,200	K, M	BX	"
CDR01BX272BK	2,700	K	BX	"
CDR01BX332B	3,300	K, M	BX	"
CDR01BX392AK	3,900	K	BX	"
CDR01BX472A	4,700	K, M	BX	"
CDR02BP221B	220	J, K	BP	"

See footnote at end of table.

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TABLE I. CDR01, CDR02, CDR03, and CDR04 capacitor characteristics - Continued.

Type designation 1/	Capacitance	Capacitance tolerance	Rated temperature and voltage temperature limits	Rated voltage
	pF			volts, dc
CDR02BP271BJ	270	J	BP	100
CDR02BX392BK	3,900	K	BX	50
CDR02BX472B	4,700	K, M	BX	50
CDR02BX562BK	5,600	K	BX	100
CDR02BX682B	6,800	K, M	BX	"
CDR02BX822BK	8,200	K	BX	"
CDR02BX103B	10,000	K, M	BX	"
CDR02BX123AK	12,000	K	BX	50
CDR02BX153A	15,000	K, M	BX	"
CDR02BX183AK	18,000	K	BX	"
CDR02BX223A	22,000	K, M	BX	"
CDR03BP331B	330	J, K	BP	100
CDR03BP391BJ	390	J	BP	"
CDR03BP471B	470	J, K	BP	"
CDR03BP561BJ	560	J	BP	"
CDR03BP681B	680	J, K	BP	"
CDR03BP821BJ	820	J	BP	"
CDR03BP102B	1,000	J, K	BP	"
CDR03BX123BK	12,000	K	BX	"
CDR03BX153B	15,000	K, M	BX	"
CDR03BX183BK	18,000	K	BX	"
CDR03BX223B	22,000	K, M	BX	"
CDR03BX273BK	27,000	K	BX	"
CDR03BX333B	33,000	K, M	BX	"
CDR03BX393AK	39,000	K	BX	50
CDR03BX473A	47,000	K, M	BX	"
CDR03BX563AK	56,000	K	BX	"
CDR03BX683A	68,000	K, M	BX	"
CDR04BP122BJ	1,200	J	BP	100
CDR04BP152B	1,500	J, K	BP	"
CDR04BP182BJ	1,800	J	BP	"
CDR04BP222B	2,200	J, K	BP	"
CDR04BP272BJ	2,700	J	BP	"
CDR04BP332B	3,300	J, K	BP	"
CDR04BX393BK	39,000	K	BX	"
CDR04BX473B	47,000	K, M	BX	"
CDR04BX563BK	56,000	K	BX	"
CDR04BX823AK	82,000	K	BX	50
CDR04BX104A	100,000	K, M	BX	"
CDR04BX124AK	120,000	K	BX	"
CDR04BX154A	150,000	K, M	BX	"
CDR04BX184AK	180,000	K	BX	"

1/ Complete type designation shall include additional symbols to indicate the voltage-temperature characteristic, capacitance tolerance (as required), termination finish, and failure rate level.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - AS, OS
Air Force - 11, 17
DLA - ES

User activities:
Navy - MC, SH
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5910-1702)

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